Sample Paper with Solutions

Module:	Software Engineering		
Module Code	EBU6304	Paper	Sample
Time allowed	2hrs	Filename	**Sample**
Rubric	ANSWER ALL FOUR QUESTIONS		

Important Note:

This sample paper is intended to help you become familiar with the structure, format and style of the exam. While it provides valuable practice, remember that actual exam questions may vary in complexity and approach.

Sample

EBU6304 Sample

Question 1

a) One of the problems with self-organising teams is that more experienced team members tend to dominate discussions and therefore to influence the team's way of working. Suggest two ways to solve this problem.

[6 marks]

	Do not write in this column
Two ways in which this problem might be reduced:	
1. All team-members propose solutions and issues anonymously on post-it notes and these are all discussed by the team. [3 marks]	
2. More junior members of the team are asked to lead discussions on issues rather than always relying on senior team members. [3 marks]	
Please note: this is an sample answer, any other sensible answer is accepted.	
	6 marks

b) Explain why estimating a user story using actual "hours" or "days" can lead to significant variations between the estimated effort and the actual effort. Suggest an estimating method to solve this problem with explanation.

[6 marks]

	Do not write in this column
Team members have differing ability and experience. E.g. What one team member might complete in 10 hours, might take another 20 hours. Senior members may underestimate the time needed by more junior members to finish the work. [2 marks]	
Using story points (or T-shirt size) may solve this problem, [2 marks] because it is an arbitrary measure (relative measure) that allows the teams to understand the size of the effort. [2 marks]	
Please note: this is an sample answer, any other sensible answer is accepted.	
	6 marks

Sample

c) What is *low-fidelity prototyping*? What are the benefits of using *low-fidelity prototyping*?

[7 marks]

	Do not write in this column
Low-fidelity prototyping is to use paper and sketch to produce prototype. [2 marks] It	
is easy to use because there is no technical learning curve [2 marks]. It can produce	
prototype quickly [1 mark]. It can be used to gather relevant feedback on early concepts [1 mark]. It is effective, low investment of time and resources. [1 mark]	
Please note: this is an sample answer, any other sensible answer is accepted.	
	7 marks

d) What is *unit testing*? Explain why it is essential to have a *refactoring* stage in the test driven development process in *unit testing*.

[6 marks]

	Do not write in this column
Unit testing tests a single component (unit) such as a function or a method defined as	
part of an object. [2 marks]	
Test driven development involves developing the code in small independent	
increments and this tends to lead to code duplication and poorly structured code. [2	
marks] Regular refactoring is essential to improve the code structure so that it is more	
readable (clean) and easy to change. [2 marks]	
Please note: this is an sample answer, any other sensible answer is accepted.	
	6 marks

Question marking: $\frac{}{6} + \frac{}{6} + \frac{}{7} + \frac{}{6} = \frac{}{25}$

Sample

Question 2

a) Answer the questions below about *Analysis* and *Design*:

[13 marks]

i) Briefly discuss what are the THREE basic stereotypes of analysis classes AND what is the function of each of them.

(6 marks)

ii) Consider the UML class diagram in Figure 1 describing a simple user management system. For each of the classes in Figure 1, identify what stereotype it belongs to.

(5 marks)

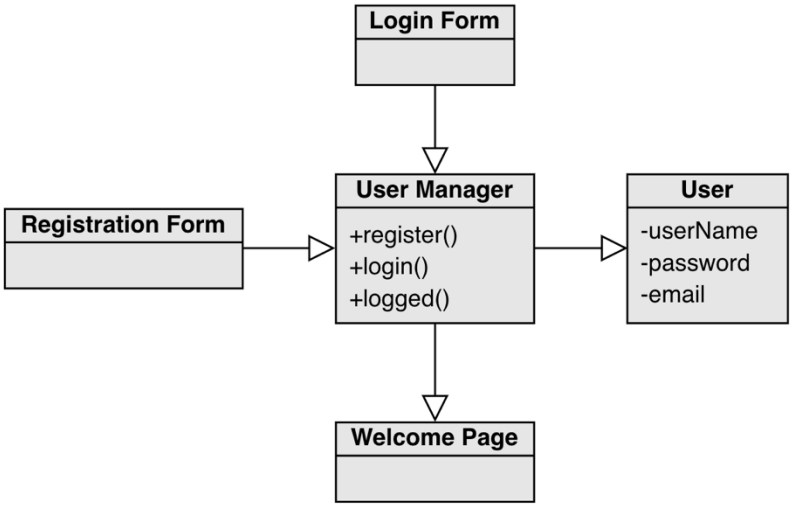


Figure 1

iii) Explain why class inheritance helps to minimise the duplication of code.

(2 marks)

	Do not write in this column
i) The three stereotypes are entity [1 mark], boundary [1 mark], and	
control [1 mark] classes. Entity classes are used to model information that is	
long-lived and persistent (such as data stored in a database) [1 mark]. Boundary	
classes model the interaction (such as user interfaces) [1 mark]. Control classes	
deal with performing tasks, getting/setting data and coordinating behaviour [1 mark].	
ii) Registration form: boundary class [1 mark]	

Login form: boundary class [1 mark]	
Welcome page: boundary class [1 mark]	
User manager: control class [1 mark]	
User: entity class [1 mark]	
iii) Thanks to inheritance child classes inherit everything from the parent [1 mark],	
so there's no need to write duplicated code unless we want to override certain	
attributes and operations [1 mark].	
Please note: this is an sample answer, any other sensible answer is accepted.	
	13 marks

b) Answer the questions below about *Software Architecture*:

[12 marks]

i) It is often said that "*non-functional requirements are a result of the architectural choices*". Discuss an example of an architectural choice in the context of web-based applications that makes the system more robust to hardware failures AND why.

(2 marks)

ii) Your company is developing a web-based application and your team leader asked you to ensure that the application is RESTful, i.e., that it respects the REST architectural constraints. In the current version of the system, even after a user is logged in, the client is still required to send its login credentials every time it makes a request to the server. Which of the REST constraints does this behaviour follow or violate AND why?

(4 marks)

iii) Fill in the blanks: "Architectural _____ can be seen as ways of representing, sharing and reusing _____."

(2 marks)

iv) What is load balancing in the context of distributed systems AND what is its aim?

(2 marks)

v) What is the business layer in the context of mobile applications architecture AND how can it be used to reduce the computational load on the mobile?

(2 marks)

Sample

	Do not write in this column
i) Replication [1 mark], of the servers can ensure robustness against hardware	
failures, because if a server stops workings others can still function [1 mark]	
Note: answers mentioning using load balancing or relying on a cloud/serverless	
model (with appropriate justifications), will also be marked as correct.	
ii) This system follows [1 mark] the statelessness constraint [1 mark], because	
according to this constraint the server should not store any state about the client	
session on the server side [1 mark] and thus when the client makes an HTTP request,	
it needs to include all information necessary for the server to fulfil	
the request [1 mark].	
iii) patterns [1 mark] *Note: "styles" is also an accepted answer*	
knowledge [1 mark]	
iv) Load balancing refers to the distribution of tasks over a set of resources [1 mark],	
with the aim of making their overall processing more efficient [1 mark].	
v) The business layer provides the main functionalities of a mobile	
application [1 mark], and it can be deployed on a remote backend to reduce the load	
on the mobile [1 mark].	
Please note: this is an sample answer, any other sensible answer is accepted.	
	12 marks

Question marking: $\frac{}{13} + \frac{}{12} = \frac{}{25}$

Sample

Question 3

a) Answer the questions below about *Project Management*:

[10 marks]

i) Describe the distinctive characteristics of software products that make project management of software projects different from other engineering projects (for example building construction).

(6 marks)

ii) In Agile project management, short daily stand-up meetings are important. Describe the purpose and activities of these daily stand-up meetings.

(4 marks)

	Do not write in this column
i) Software products are intangible (cannot be seen or touched as physical products) [1 mark] and flexible (can be changed and altered frequently at any time) [1 mark]. Software engineering is not recognised as a "sane" engineering discipline (not standardised). [1 mark] Many software projects are "one-off' projects" [1 mark] meaning: Technologies are changing too fast. [1 mark] and previous experiences may be obsolete. [1 mark]	
NOTE: The explanations can be different but it is just to establish that the student understands the concepts. If a student mention "require a perceptive insight" as one of the answer, it is accepted.	
ii) The purpose of the daily stand-up meetings is that everyone on the team knows what is going on and, if problems arise, can re-plan short-term work to cope with them. [1 mark]	
The activities that take place at daily stand-up meetings are:	
All team members share information. [1 mark]	
They describe their progress since the last meeting. [1 mark]	
They discuss problems that have arisen and what is planned for the following day. [1 mark]	
Please note: this is an sample answer, any other sensible answer is accepted.	
	10 marks

Sample

b) Alpha & Beta is a small IT company that has been providing PC and desktop-based applications to its customers for the past 10 years. Due to the current COVID-19 pandemic, the company immediately moved all its applications to the Cloud as many customers work online remotely. Because of this sudden move to the Cloud, many customers are complaining that some of the software that used to work well on their PCs and desktops are not working properly and there is no alternative for a PC and desktop-based applications for them to use. In addition, the company does not have staff with the right skills to manage the Cloud-based applications as this is new to them. Answer the following questions.

[10 marks]

i) Briefly discuss the problems that Alpha & Beta company is facing as a result of the sudden move to the Cloud-based applications due to COVID-19 pandemic (use professional software engineering terminologies).

(3 marks)

ii) Discuss what Alpha & Beta IT company failed to do which could have prevented the problem they are now facing.

(3 marks)

iii) Suggest two solutions to the company on how to how to improve their management.

(4 marks)

	Do not write in this column
i) Project planning problems: The COVID-19 pandemic shows that we cannot plan for everything that can go wrong in software engineering projects and that projects do not always follow as we planned [1 mark].	
There was no proper planning for alternative or contingency plans to fall back to the PC and desktop-based applications in case the cloud-based applications fail [1 mark].	
In addition, the company does not have staff with cloud-based application skills which is a problem of poor risk management on the part of the company [1 mark].	
ii) The company failed to make alternative plans or contingency plans to revert to the PC or desktop-based applications in case of any problems with cloud-based applications [1 mark].	
The company did not apply Avoidance Strategy before choosing to move completely to the cloud as they need to be careful about using new or unfamiliar products (tools, software, hardware, technology (in this case cloud)) to balance any benefit obtained from risk they will not perform as expected [1 mark].	
The company did not train its staff or hire staff with cloud-based applications skills before moving it entire applications to the cloud [1 mark].	

Sample

iii)The company should first get back to its PC and desktop-based applications and allow its staff and customers to install the software on their machines [1 mark] and use online platforms like MS Teams, Zoom, etc to communicate and work remotely [1 mark].	
The company should train its staff on how to use and develop cloud-based version of its PC and desktop-based applications [1 mark] or hire new staff that have cloud-based application skills or outsource the development of cloud-based application [1 mark].	
Please note: this is an sample answer, any other sensible answer is accepted.	
	10 marks

c) Describe the four freedoms of *Free Software* and the benefit of having *Free Software.*

[5 marks]

	Do not write in this column
The freedom to run the program, for any purpose [1 mark]	
The freedom to study how the program works, and change it so it does your computing as you wish and access to the source code is a precondition for this. [1 mark]	
The freedom to redistribute copies so you can help your neighbour. [1 mark])	
The freedom to distribute copies of your modified versions to others [1 mark]. By doing this, you can give the whole community a chance to benefit from your changes and access to the source code is a precondition for this. [1 mark]	
Please note: this is an sample answer, any other sensible answer is accepted.	
	5 marks

Question marking: $\frac{}{10} + \frac{}{10} + \frac{}{5} = \frac{}{25}$

Sample

Question 4

A technology company has been using a software product for managing its graduate recruitment and associated management-training programme for many years. Now the company decides to use Artificial Intelligence (AI) software to improve the success rate for graduates progressing to become senior managers in the company. Data used to train this AI system is based on all the graduates who have joined the company previously. Answer the following questions:

a) Explain why the AI system as adopted by the company might be inherently biased.

[3 marks]

	Do not write in this column
The data has been trained using a skewed sample. For example, statistically, more men than woman have applied to join tech firms historically, so the proportion of male graduates in the company is likely to be higher than that of female ones [2 marks – for either stating 'skewed sample' or giving the actual example]. This means the AI system will be more biased [for example, towards male attributes than females ones] [1 mark].	
Please note: this is an sample answer, any other sensible answer is accepted.	
	3 marks

b) In the context of 'fairness' in the AI system, explain the difference between statistical parity and predictive equality.

[8 marks]

	Do not write in this column
Statistical parity is a **measure of the ratio** of the probability of a positive outcome for one group relative to the probability of a positive outcome in another group [2 marks]. For 'fairness' in recruitment (as used in the context of the AI system), the ratio (or percentage) should **be the same as the underlying population** [2 marks]. [So for example, if 100 males and 40 females apply join the company, there would be statistical parity only if 10 females get accepted and 30 get rejected if 25 males get accepted and 75 get rejected, i.e. equal ratio of 1:3 between accepted:rejected]. *Predictive equality*, on the other hand, is when the accuracy of decisions made by the system is **equal across two different groups** [2 marks], as measured by a **false positive error rate** [or conditional accuracy] [2 marks]. [In other words, the AI should classify risk based only on features that truly matter, so there should be no inherent gender bias.]	
Please note: this is an sample answer, any other sensible answer is accepted.	
	8 marks

c) In general terms, what important ethics principle should be applied when collecting data from the graduates to inform the AI system? In giving your answer, reference should be made to the information that the graduates should be provided with, and where relevant, to the provisions of the UK Data Protection Act (DPA) 2018 or the European Union's General Data Protection Regulation (GDPR).

[14 marks]

	Do not write in this column
It is important that the graduates give their '**informed consent**' [1 mark] before any data is collected from them. This consent could be obtained by **providing them with information** [e.g. through a participant information sheet or other handout] that **explains**: • **Why the information is being collected** [1 mark] [i.e. purpose for which the data will be used and the context]. • **Why they have been chosen** or selected to take part [1 mark], and **What would happen if they agree to take part** in the data collection [1 mark] [e.g. including any further surveys or research planned]. • Whether there are **any benefits [**or inducements] for them to take part [1 mark] [i.e. participations should ideally be reimbursed for their time/data where possible]. • Whether there are **any risks involved** [1 mark]. • How their participation (and data collected) **will be kept confidential**. [1 mark] This data is to be held and used in accordance with the **DPA** 1998 [instead of DPA, they can also say GDPR – either will suffice] [1 mark] • Who **they may contact** for **further information** [1 mark] or **if things go wrong** [1 mark] • They should be told that they may **withdraw consent** [1 mark] at any time **without giving a reason** [1 mark] Once a withdrawal request has been made, they have the '**right to be forgotten**' [1 mark] according to **DPA/GDPR** [1 mark].	
Please note: this is an sample answer, any other sensible answer is accepted.	
	14 marks

Question marking: $\frac{}{3} + \frac{}{8} + \frac{}{14} = \frac{}{25}$